

①

$$y = x^2 + 1$$
$$y^2 = x + 1$$

let $x = 5$

let $x = 4$

$$y = 5^2 + 1 = 26$$

$y = 26$ is mean ~~that~~ Function

$$y^2 = 4 + 1 = 9 \quad y^2 = 9$$

$$y = \sqrt{9}$$

$$y = 3$$

or $y = -3$ not function

②

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(n) = 3n$$

One input gives only one output

$$g: \{1, 2, 3\} \rightarrow \{a, b, c\} \text{ defined by}$$

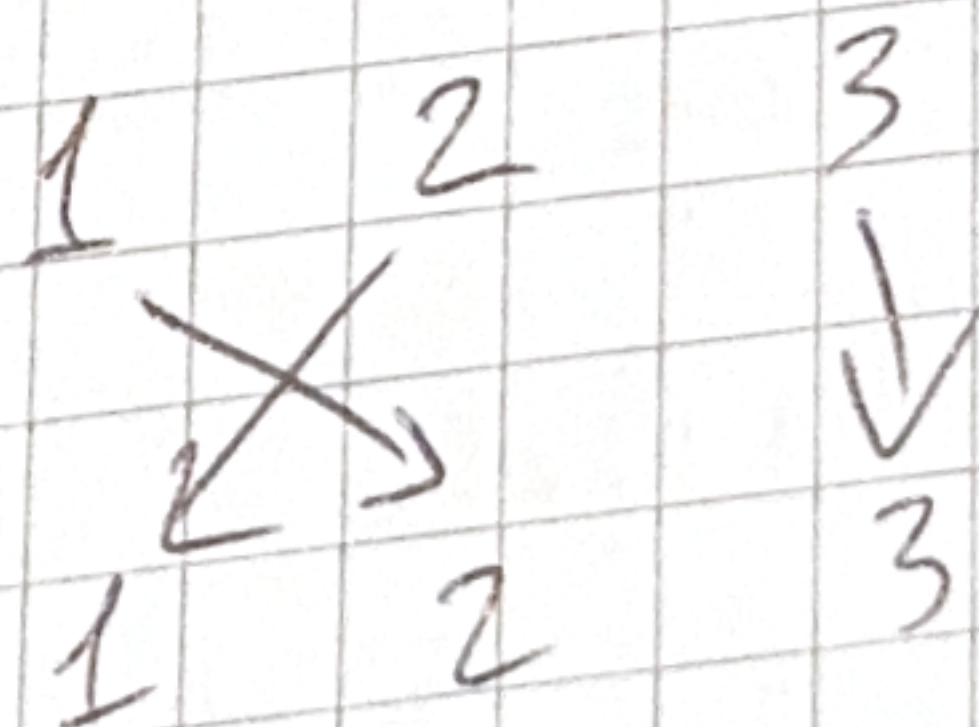
$$g = \begin{pmatrix} 1 & 2 & 3 \\ c & a & a \end{pmatrix}$$

g isn't surjective "b" is missing

$$h: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$$

is surjective because

each output has at least one input



③ $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined $f(n) = 3n$

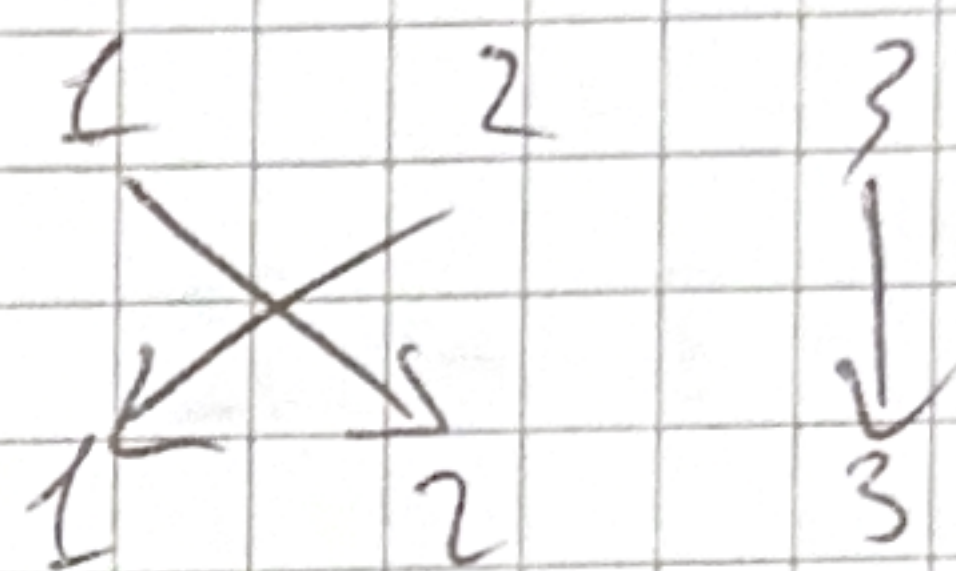
f is injective because one integer gives one single result

$$g: \{1, 2, 3\} \rightarrow \{a, b, c\} \quad g = \begin{pmatrix} 1 & 2 & 3 \\ c & a & a \end{pmatrix}$$

g isn't injective because two inputs assigned to one output

$$h: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$$

h is injective each



output has only one input

④ if $f(x) = \frac{1}{x+2}$ and $g(x) = \frac{1}{x} - 2$

is $g = f^{-1}$?

$$g(f(x)) = \frac{1}{\left(\frac{1}{x+2}\right) - 2} = x + 2 - 2 = x$$

so $g = f^{-1}$ and $f = g^{-1}$

⑤ Inverse function $f(x) = 2 + \sqrt{x-4}$

$$y = 2 + \sqrt{x-4} \quad (y-2)^2 = x-4 \quad x = (y-2)^2 + 4$$

Domain of f is $[4, +\infty)$

The range of f is $[2, +\infty)$

f^{-1} is $[2, +\infty)$

⑥ Inverse function $C = \frac{5}{9}(F - 32)$

$$C = \frac{5}{9}(F - 32) = C \cdot \frac{9}{5} = F - 32 \Rightarrow$$

$$= F = \frac{9C}{5} + 32$$

$$C = h(F) = \frac{5}{9}(F - 32) \quad F = h^{-1}(C) = \frac{9C}{5} + 32$$

⑦ Find domain and range $g(x) = 2\sqrt{x-4}$

~~$y = 2$~~ $x - 4 \geq 0 \Rightarrow x \geq 4$

Domain $x \in [4, +\infty)$

Range $g(x) \in [0, +\infty)$

⑧ Find domain and range $h(x) = -2x^2 + 4x - 9$

$$y = ax^2 + bx + c \quad a = -2, \quad b = 4 \quad -b/2a$$

$$f\left(\frac{-b}{2a}\right) = \cancel{w} - \frac{4}{2 \cdot (-2)} = -1 \quad x = -1$$

$$h(x) = -2 \cdot (-1) + 4 \cdot (-1) - 9 = 2 - 4 - 9 = -11$$

$$R: [-11, +\infty)$$

⑨ Find domain $f(x) = \frac{x-4}{x^2-2x-15}$

$$x^2 - 2x - 15 = 0$$

$$(x-5)(x+3) = 0$$

$$x+3=0$$

$$x \quad \times \quad -5$$

$$x-5=0$$

$$x=5$$

$$x=-3$$

$$f(x) = x - 4x^2 - 2x - 15$$

$$f(x) \text{ is } (-\infty, -3) \cup (-3, -5) \cup (-5, +\infty)$$